

Semester	B.E. Semester VII – CMPN
Subject	DevOps
Subject Professor In-charge	Dr. Snehal Andhare
Assisting Teachers	Prof. Snehal Andhare
Classroom	M414A

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Grade and Subject Teacher's Signature	

Practical Number	5
Practical Title	To understand Jenkins Master-Slave Architecture and scale your Jenkins standalone implementation by implementing slave nodes.
Objectives (Skill Set / Knowledge Tested / Imparted)	1. To understand the Master-Slave (Controller-Agent) architecture in Jenkins. 2. To configure and connect a Jenkins agent node with the Jenkins controller. 3. To execute Jenkins build tasks using an agent node.
Description	The practical demonstrates scaling Jenkins by configuring a distributed Master-Slave (Controller-Agent) architecture. A standalone Jenkins master manages scheduling, dispatching jobs, and monitoring builds, while an external slave node is provisioned to execute the workload. The agent node is configured with dedicated workspace directories and labeled accordingly. Upon establishing a successful remoting connection via Java Web Start/WebSocket, a build job is executed remotely on the slave agent, validating offloaded execution and distributed CI capabilities.
Program	

Output

Agent Setup:

Agent Connected:

```

Microsoft Windows [Version 10.0.26100.4351]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VIT>curl.exe -s0 http://localhost:8082/jnlpJars/agent.jar & java -jar agent.jar -url http://localhost:8082/ -se
cret cfb573f960ccc8c418a27348b62128e52f31627c36f07d491a83b44d675dd96 -name Kunal -webSocket -workDir "C:\Users\VIT\DevO
ps_Kunal\Experiment-5"
Aug 25, 2026 1:06:44 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\Users\VIT\DevOps_Kunal\Experiment-5\remoting as a remoting work directory
Aug 25, 2026 1:06:44 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to C:\Users\VIT\DevOps_Kunal\Experiment-5\remoting
Aug 25, 2026 1:06:44 PM hudson.remoting.Launcher createEngine
INFO: Setting up agent: Kunal
Aug 25, 2026 1:06:44 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3355.3357.v931d3c992987
Aug 25, 2026 1:06:44 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\Users\VIT\DevOps_Kunal\Experiment-5\remoting as a remoting work directory
Aug 25, 2026 1:06:44 PM hudson.remoting.Launcher$CuiListener status
INFO: WebSocket connection open
Aug 25, 2026 1:06:44 PM hudson.remoting.Launcher$CuiListener status
INFO: Connected
  
```

Running Task through Agent:

Conclusion

Thus, the practical demonstrated the implementation and functioning of the Jenkins Master-Slave architecture.

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